

Problems encountered with the glasses

Privacy Analysis of Augmented Reality Glasses

Pairing

The pairing between the glasses and phone appears to be quite sensitive and unstable. The following are some possible encounters:

The phone is not able to find and connect the glasses

Possible reasons:

- When glasses and/case are low on battery
Try: Powering off the glasses and place them into the case and charge until either the light on the case turns green or for at least an hour.
- When glasses were not in case for some time
Try: Placing the glasses in the case for at least half an hour
- When the glasses are already paired to another device
Try: Factor reset the glasses.
The glasses can only be paired with one device at a time so one will need to factory reset the glasses before another can pair the glasses to their account.
The instructions for factory reset can be found below in the section "Instructions".
- If it is not because of any of the above reasons:
Try: Unpair device, and re-pair the glasses.
The following should be on the Meta View App when you try to add a device (similar to how you re-pair the device).
Power on the glasses and put them back into the charging case.
Push the button on the back till it turns blue and then connect the device.

The progress bar stops completely whilst importing media captures

Possible reasons:

- When you do not have an internet
- When the glasses are not connected
- None of the above

This is for case No. 3 where you have internet. Furthermore, in settings the glasses still show to be paired in Bluetooth as well as on the top left corner of the Meta View app. The top left corner shows the Bluetooth symbol alongside the battery percentage if connected to the glasses and shows three wriggling dots when not connected. However, the photos/videos are not able to be imported from the glasses despite connected to the phone. The Meta View app should look something like this when trying to import the media captures:

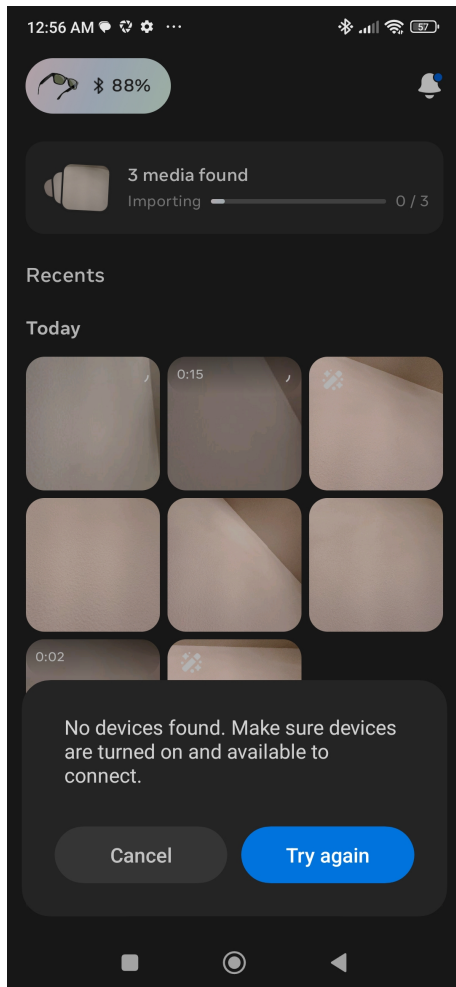


Figure 1: The progress bar does not move and the app prompts that there is no device although the top left corner shows it is connected

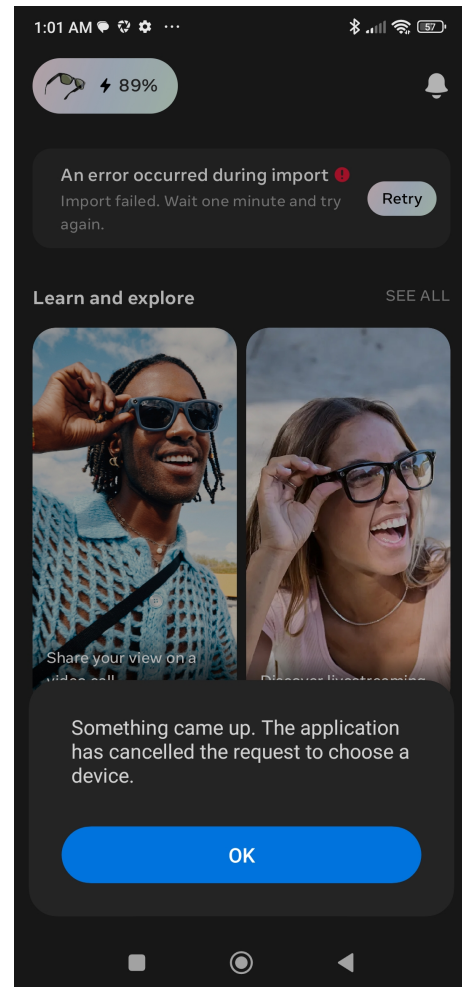


Figure 2: An error occurs and the app prompts that there was an error whilst the device is connected and retry does not work or leads to Figure 1

Try:

- Method 1: Reboot the phone
Reboot the phone.
Open the Meta app.
Power on and wear the glasses and wait for a few minutes.
- Method 2: Unpair and Re-pair the glasses
The instructions to unpairing and re-pairing the glasses can be found below in the section "Instructions".
- Method 3: Factory reset the glasses
This is more extreme as it will delete all the photos/videos saved on the glasses.
Make sure that if you do factory reset that you also unpair the device on your phone first.
The instructions to unpairing and re-pairing the glasses can be found below in the section "Instructions".
The instructions for factory reset can be found below in the section "Instructions".
- Method 4: Uninstall and Reinstall the app
This will delete all media on both the phone and the glasses as it will require you to factory reset the glasses, so make sure to save any photos you want to keep to the photo album on the phone. After reinstalling the app, pair the glasses as you did the first time (following the instructions on the app) though it may prompt you to forget (i.e. unpair the glasses in Bluetooth settings) and factory reset the glasses. The instructions to factory reset the glasses can be found below in the section "Instructions".

Privacy Breach

- Messages
One can ask Meta whether one has received new messages. Meta will read out the name of the person who sent you the message and read the message out loud.
Note that this can be done whilst the phone is closed and since there is no voice recognition, anyone who has the glasses is able to command the AI on the glasses to read the received messages.
- Indicator LED
Although there is a light to indicate to others when one is taking photos/videos. There is a way to bypass this by first starting the video recording then cover up the lights. This does not trigger the security warning.
- Photos/Videos being used to train AI models.
In the app you are able to choose the following options in regards to data sharing.

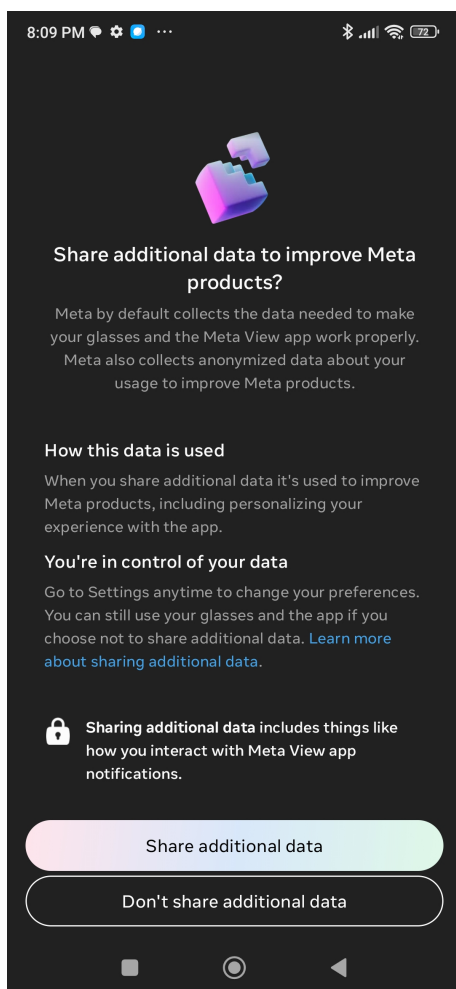


Figure 3: Share additional data

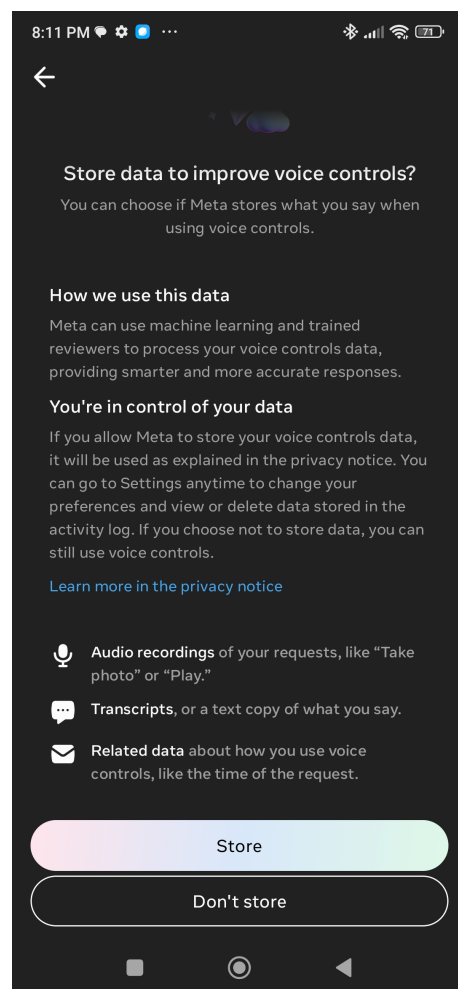


Figure 4: Share voice recordings

However, there is a third in regards to images that you share is being processed by Meta to their AI models. This is especially unfortunate for people living in the US as they do not have the ability to say no. In EU and UK, one is supposed to be able to apply to opt out of this where you need to state your reasoning, and more. But there has been alleged cases where Meta rejected the request.

Instructions

Unpairing the glasses

Either: In Settings,

1. Go to Bluetooth and unpair the glasses;

Or: In the Meta View app,

1. press settings at the bottom right.
2. Tap 'Your glasses' or directly the device you want to unpair.
3. Tap 'General', 'Unpair glasses', 'Unpair'.

Note that the first option is more preferred and easier.

Re-pairing the glasses

1. Open the Meta View app.
2. Go to the settings at the bottom right and press 'Add Device' on the top corner of the app.
3. Choose the Meta Ray Ban glasses.
4. Take the glasses out of the case and power on.
5. Put the glasses back inside the case and close the case.
6. Wait a bit for the LED on the case to start blinking blue, this means that the device is ready to pair with the Meta View app. If the LED does not blink blue, press and hold the button on the back of the case for at least 5 seconds till you see blinking blue.
7. On the Meta View app, press settings at the bottom right and then follow the on-screen instructions.

Factory Reset

1. Power on the glasses and only need to close the right arm.
2. Press the capture button 5 times very quickly then the notification LED should turn orange.
3. Press and hold the capture button until that notificaiton LED starts blinking.
4. Then release the capture button wait for the LED to turn blue that indicates that you can now pair to a new device.